1. Notes on Multiplexer and Decoder

Multiplexer (MUX)

A multiplexer is a digital circuit i which we selects one input from several inputs and sends the selected input to the output. It is also called a data selector.

we are using a 4-to-1 multiplexer. It has four data inputs: D0, D1, D2 and D3, two select inputs: S1 and S0, and one output Y.

The two select inputs decide which data input will be connected to the output.

- S1 S0 = 00 → D0 is selected
- S1 S0 = 01 → D1 is selected
- S1 S0 = 10 → D2 is selected
- S1 S0 = 11 → D3 is selected

So, the main purpose of a multiplexer is to choose one input and send it to the output.

Decoder

A decoder is a digital circuit which converts a binary input into one active output. A 2-to-4 decoder has two input lines and four output lines.

The two inputs are A and B, while the outputs are Y0, Y1, Y2 and Y3. For every input combination, only one output becomes active.

- AB = 00 → Y0 = 1
- AB = 01 → Y1 = 1
- AB = 10 → Y2 = 1
- AB = 11 → Y3 = 1

The decoder is useful when we need to select one output based on a binary input.

2. Solved Examples

Example 1: 4-to-1 Multiplexer

Given: D0 = 1, D1 = 0, D2 = 1, D3 = 0, S1 = 1, S0 = 0

Solution: S1 S0 = 10. According to the MUX selection, 10 → D2. D2 = 1. Therefore, Y = 1.

Answer: Output Y = 1

Example 2: 2-to-4 Decoder

Given: A = 1, B = 0

Solution: AB = 10. For a 2-to-4 decoder, 10 → Y2.

Therefore: Y0 = 0, Y1 = 0, Y2 = 1, Y3 = 0.

Answer: Y2 = 1

3. Pseudo-code of Main Function

MUX Function

- Take the values of D0, D1, D2 and D3.
- Take the values of S1 and S0.
- Use S1 and S0 to find the selected input.
- If S1S0 is 00, select D0.
- If S1S0 is 01, select D1.
- If S1S0 is 10, select D2.
- If S1S0 is 11, select D3.
- Show the selected value as the output Y.

Decoder Function

- Take the two input values A and B.
- Check the binary combination of A and B.
- If AB is 00, activate Y0.
- If AB is 01, activate Y1.

- If AB is 10, activate Y2.
- If AB is 11, activate Y3.
- Keep the other outputs at 0.
- Display the result.